INFORMATICS PRACTICES PROJECT

EXPENSE TRACKER SYSTEM USING PYTHON & MYSQL



Submitted By: Harshit Singh

Class: XII Section: Comm.B

Roll Number: 12

Submitted To: Ms.Aakanksha Rikhari

Academic Session: 2026 – 27

Subject: Informatics Practices (IP)

CERTIFICATE

This is to certify that Harshit Singh of Class XII Comm. B has successfully completed the Informatics Practices project titled "Expense Tracker System Using Python & MySQL" during the academic session 2026–27 under the guidance of the subject teacher.

The project submitted is a genuine work carried out by the student according to the guidelines prescribed by CBSE. The student has shown sincere effort, dedication, and understanding while completing the project work.

We appreciate the hard work and successful completion of this project.

Teacher's Signature: _____

Date: 01.06.2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my Informatics Practices teacher for providing guidance and support throughout the completion of this project. Their valuable suggestions and encouragement helped me understand various programming concepts and database management techniques.

I am also thankful to my school for providing the necessary resources and environment required for the successful completion of this project. I would like to thank my parents and friends for their continuous support, motivation, and encouragement during the preparation of the project.

This project helped me improve my knowledge of Python programming, MySQL database connectivity, and data visualization techniques. It also enhanced my practical understanding of real-life applications of technology.

Lastly, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

INDEX

S.No	Topic	Page No.
1	Introduction	1
2	Objective of the Project	2
3	Software Requirements	3
4	Theory / Research	4
5	Database Section (MySQL)	5
6	Coding Section (Python)	6
7	Output Screenshots	7
8	Data Visualization	8
9	Conclusion	9
10	Bibliography	10

INTRODUCTION

In today's digital world, managing personal expenses has become very important. Many people face difficulty in keeping records of their daily spending manually. An Expense Tracker System helps users store, manage, and analyze their expenses efficiently using technology. This project is developed using Python and MySQL to simplify expense management and provide accurate financial records.

The Expense Tracker System allows users to add, view, search, and analyze their expenses through a user-friendly program. All records are stored in a MySQL database, which helps in maintaining data securely and systematically. The project also uses Python libraries such as mysql.connector, pandas, and matplotlib for database connectivity, data handling, and graphical representation.

The main purpose of this project is to reduce manual calculations and improve financial management skills. Users can categorize expenses such as food, transport, shopping, education, and entertainment. The system can generate reports and charts that help users understand their spending habits.

This project demonstrates the practical implementation of programming concepts like loops, conditional statements, functions, file/database handling, and data visualization. It also improves analytical thinking and problem-solving abilities. The Expense Tracker System is useful for students, employees, and families who want to maintain a proper record of their expenses and manage their budget effectively.

OBJECTIVE

The main objective of the Expense Tracker System is to help users manage and monitor their daily expenses in an organized manner. The project is designed to reduce the difficulties of maintaining manual expense records and to provide a simple digital solution using Python and MySQL.

This project helps users store expense details such as amount, category, date, and description in a MySQL database. Users can easily add new expenses, search previous records, and view complete expense reports whenever required. The system also provides graphical analysis of expenses through charts and graphs, making financial analysis easier and more understandable.

Another important objective of this project is to apply programming concepts learned in Informatics Practices to real-life applications. The project uses concepts like loops, functions, conditional statements, database connectivity, and data visualization. It also improves logical thinking, problem-solving ability, and practical coding skills.

The Expense Tracker System aims to promote better financial management habits among users. By tracking expenses regularly, users can identify unnecessary spending and plan their budgets more effectively. The project also demonstrates how technology can simplify everyday life activities through automation and digital record management.

SOFTWARE REQUIREMENTS

The Expense Tracker System is developed using Python programming language and MySQL database. Various software tools and libraries are used to build and run the project successfully. These software requirements help in database management, coding, data analysis, and graphical representation of expense records.

The main programming language used in this project is Python because it is simple, powerful, and easy to understand. Python provides various built-in functions and libraries that make application development faster and more efficient. The project is executed using an IDE such as IDLE, VS Code, or PyCharm.

MySQL is used as the backend database to store expense details securely. It helps in maintaining records systematically and allows users to retrieve and manage data easily. The connection between Python and MySQL is established using the mysql.connector library.

Additional libraries such as pandas and matplotlib are also used in the project. Pandas helps in organizing and handling data efficiently, while Matplotlib is used to create graphs and charts for expense analysis.

Software and Libraries Used

Software / Library	Purpose
Python	**Main programming language**
MySQL	**Database management**
mysql.connector	**Connect Python with MySQL**
pandas	**Data handling and analysis**
matplotlib	**Creating charts and graphs**
VS Code / IDLE	**Writing and running code**

THEORY / RESEARCH

An Expense Tracker System is a software application used to record, manage, and analyze daily expenses in a systematic manner. It helps users maintain financial records digitally instead of using traditional paper-based methods. With the increasing use of technology, digital expense management systems have become more efficient, accurate, and user-friendly.

The Expense Tracker System developed in this project is based on Python programming and MySQL database management. Python is widely used for application development because of its simple syntax and powerful libraries. MySQL is used to store expense data securely and efficiently. The combination of Python and MySQL provides a reliable platform for handling large amounts of financial records.

The system stores important information such as expense amount, category, date, and description. Users can perform operations like adding new expenses, viewing all records, searching expenses, and generating reports. The project also includes graphical representation of data using charts and graphs. These visualizations help users understand their spending patterns and financial habits more clearly.

The project uses various programming concepts such as loops, conditional statements, functions, database connectivity, and data visualization. Libraries like mysql.connector, pandas, and matplotlib are used to improve functionality and efficiency.

This system is beneficial for students, employees, shopkeepers, and families who want to maintain proper financial records and manage their budgets effectively. It reduces manual work, minimizes errors, and saves time through automation and organized data management.

DATABASE SECTION (MySQL)

The Expense Tracker System uses a MySQL database to store and manage expense records efficiently. A database helps in organizing data systematically and allows users to retrieve, update, and analyze records whenever required. In this project, MySQL is connected with Python using the mysql.connector library.

The database stores important details such as Expense ID, Expense Category, Amount, Date, and Description. Each expense entered by the user is saved permanently in the database table. This makes the system more secure, reliable, and efficient compared to manual record keeping.

Database Name:

expense_tracker

Table Name:

Expenses

MySQL Table Creation Code:

```
CREATE DATABASE expense_tracker;

USE expense_tracker;

CREATE TABLE expenses (

        expense_id INT PRIMARY KEY AUTO_INCREMENT,

        category VARCHAR(50),

        amount FLOAT,

        expense_date DATE,

        description VARCHAR(100)

        );
```

Sample Records in Table:

Expense ID	Category	Amount	Date	Description
1	Food	250	2026-05-01	Lunch
2	Transport	120	2026-05-02	Bus Fare
3	Shopping	1500	2026-05-03	Clothes
4	Education	800	2026-05-04	Books

Purpose of Database:

- Stores all expense records permanently
- Helps in searching and updating data easily
- Reduces chances of data loss
- Provides fast and accurate record management
- Makes report generation easier

The MySQL database plays an important role in the Expense Tracker System by ensuring organized and secure storage of financial data.

CODING SECTION (Python)

```
import mysql.connector

import pandas as pd

import matplotlib.pyplot as plt


# MySQL Connection

con = mysql.connector.connect(

                        host="localhost",

                        user="root",

                        password="your_password",

                        database="expense_tracker"

                        )


                        cursor = con.cursor()


# Function to Add Expense

def add_expense():

            category = input("Enter Category: ")

            amount = float(input("Enter Amount: "))

            date = input("Enter Date (YYYY-MM-DD): ")

           description = input("Enter Description: ")
```

```
        query = "INSERT INTO expenses(category, amount, expense_date, description)
VALUES(%s,%s,%s,%s)"

        values = (category, amount, date, description)



        cursor.execute(query, values)

        con.commit()


print("Expense Added Successfully!")


# Function to View Expenses

def view_expenses():

    query = "SELECT * FROM expenses"

    cursor.execute(query)


    records = cursor.fetchall()


    print("\nExpense Records")

    print("-" * 50)


    for row in records:

        print(row)


# Function to Search Expense
```

```
def search_expense():
    category = input("Enter Category to Search: ")

    query = "SELECT * FROM expenses WHERE category=%s"
    value = (category,)

    cursor.execute(query, value)

    records = cursor.fetchall()

    if records:
        for row in records:
            print(row)
    else:
        print("No Record Found")

# Function for Expense Analysis Graph
def expense_graph():
    query = "SELECT category, SUM(amount) FROM expenses GROUP BY category"

    df = pd.read_sql(query, con)

    plt.bar(df['category'], df['SUM(amount)'])
```

```
    plt.xlabel("Category")
    plt.ylabel("Amount")
    plt.title("Category Wise Expenses")
    plt.show()

# Main Menu
while True:
    print("\n===== EXPENSE TRACKER SYSTEM =====")
    print("1. Add Expense")
    print("2. View Expenses")
    print("3. Search Expense")
    print("4. Expense Graph")
    print("5. Exit")

    choice = int(input("Enter Your Choice: "))

    if choice == 1:
        add_expense()

    elif choice == 2:
        view_expenses()

    elif choice == 3:
```

```
        search_expense()


    elif choice == 4:
        expense_graph()


    elif choice == 5:
        print("Thank You")
        break


    else:
        print("Invalid Choice")


con.close()
```

Explanation of the Code:

The Python program connects with the MySQL database using the mysql.connector library. Different functions are created to perform operations such as adding expenses, viewing records, searching expenses, and displaying graphical analysis.

The add_expense() function stores new expense details into the MySQL database. The view_expenses() function displays all saved records. The search_expense() function helps users find expenses according to category. The expense_graph() function uses Pandas and Matplotlib libraries to generate a bar graph for category-wise expense analysis.

The program uses loops, conditional statements, functions, and database handling concepts. A menu-driven interface is provided to make the system user-friendly and easy to operate.

OUTPUT SCREENSHOTS SECTION

The Output Screenshots section contains images of the program execution, MySQL database tables, menu options, and graphical reports generated by the Expense Tracker System. These screenshots help in understanding the working of the project and provide proof of successful execution of the program.

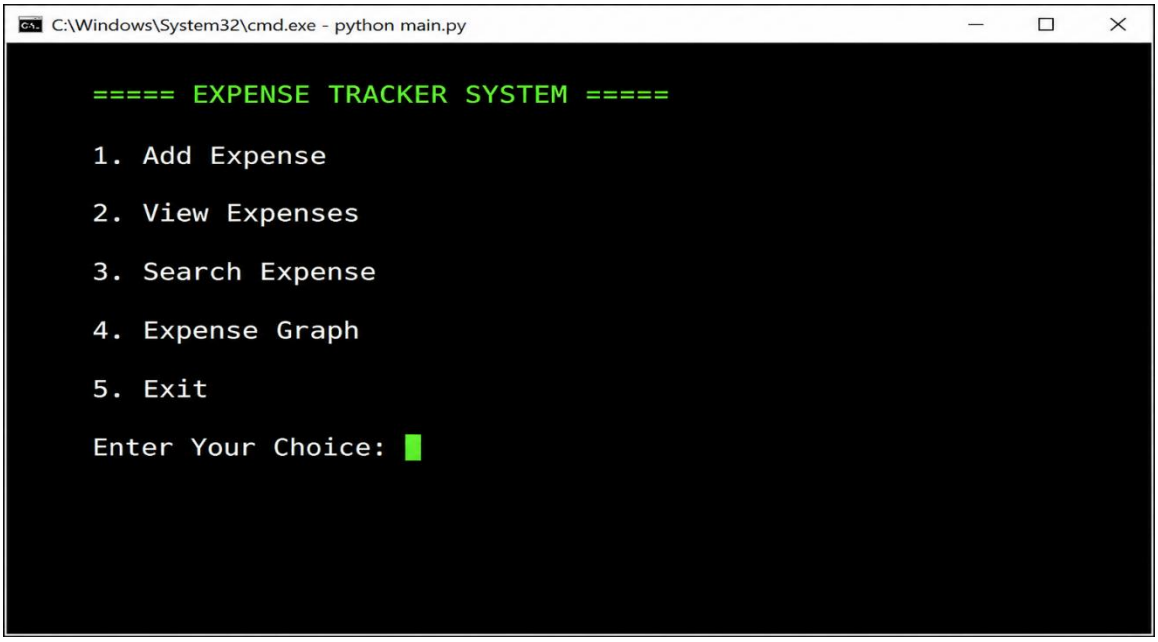
The screenshots included in this project should be clear, properly cropped, and labeled according to their function. Different outputs demonstrate how users can interact with the system to add, view, search, and analyze expense records.

Screenshots to Attach

1. Main Menu Screen

This screenshot should display the main menu of the Expense Tracker System showing all available options such as:

- Add Expense
- View Expenses
- Search Expense
- Expense Graph
- Exit



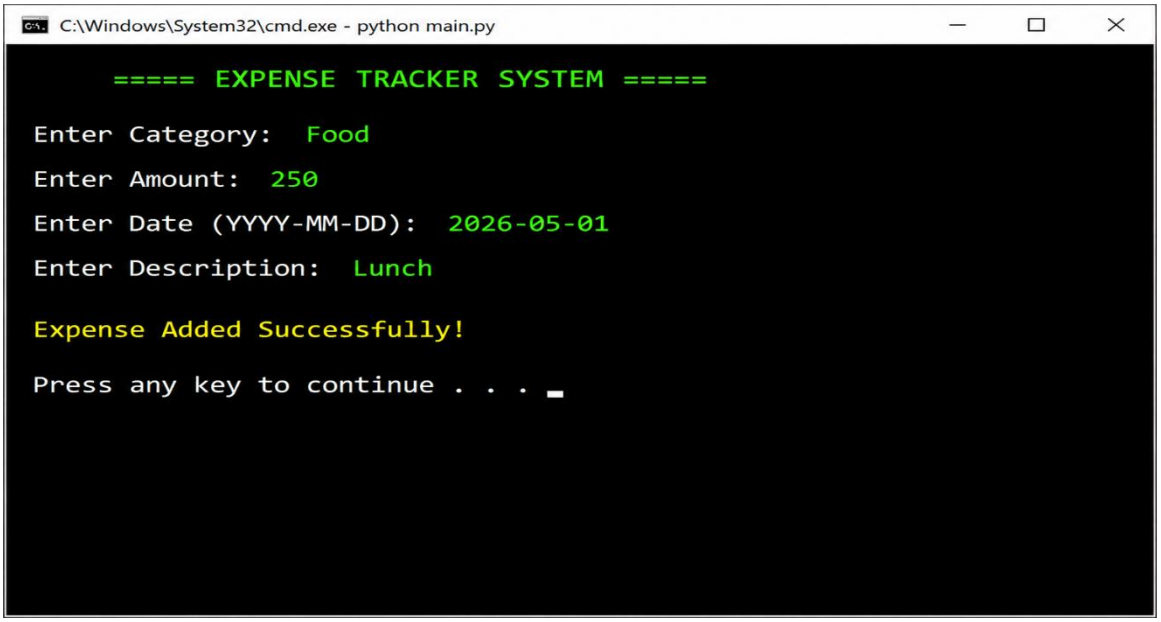
2. Add Expense Output

This screenshot should show the successful insertion of expense details entered by the user into the MySQL database.

Example:

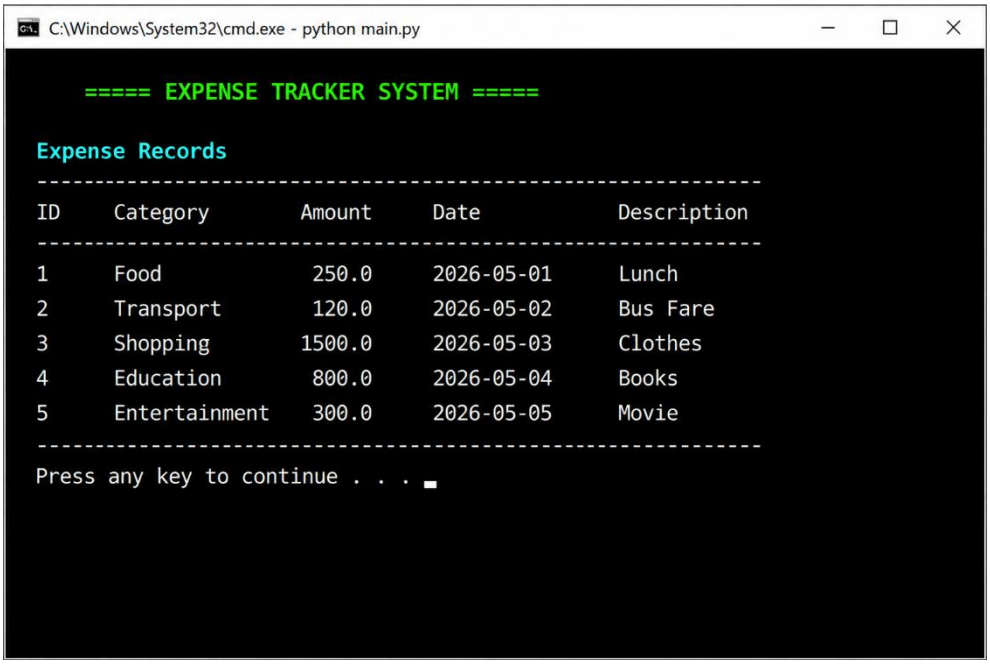
Enter Category: Food
Enter Amount: 250
Enter Date: 2026-05-01
Enter Description: Lunch

Expense Added Successfully!



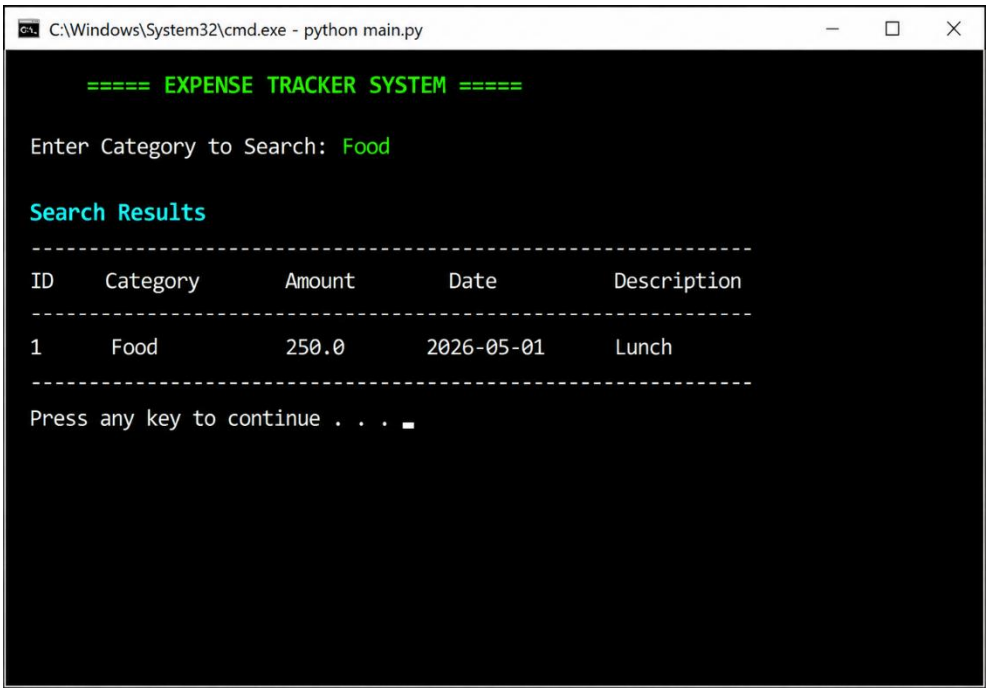
3. View Expenses Output

This screenshot should display all expense records stored in the database table.



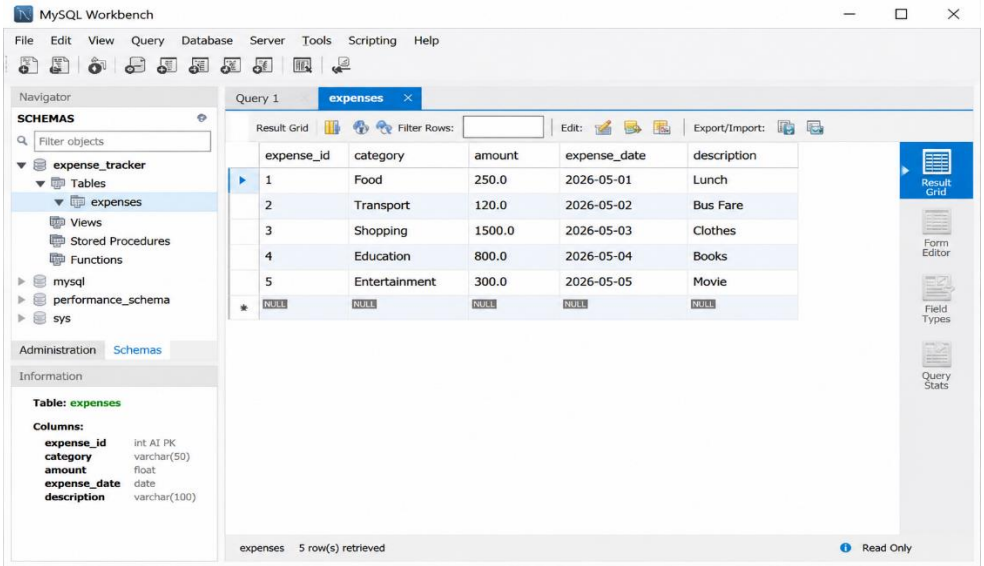
4. Search Expense Output

This screenshot should show how the program searches and displays records based on a category entered by the user.



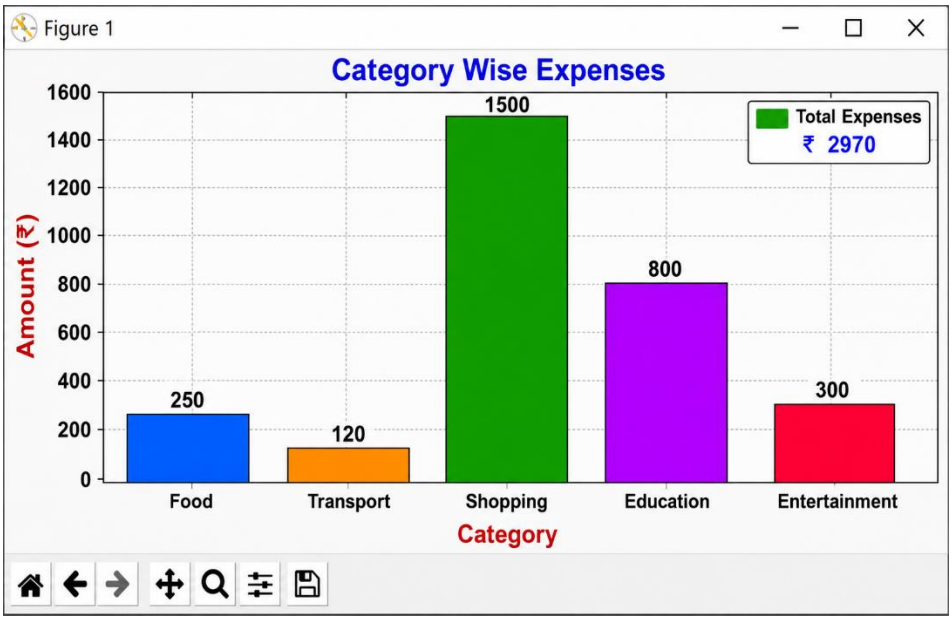
5. MySQL Database Screenshot

Attach a screenshot of the expenses table from MySQL showing saved records.



6. Expense Graph Screenshot

Attach the bar graph or pie chart generated using Matplotlib for expense analysis.



Importance of Output Screenshots

- Proves successful execution of the project .
- Helps in understanding program functionality .
- Makes the project report more practical and attractive .
- Demonstrates database connectivity and graphical analysis .
- Improves documentation quality and presentation.

DATA VISUALIZATION

Data visualization is the graphical representation of data using charts and graphs. In the Expense Tracker System, data visualization helps users understand their spending patterns in a simple and effective manner. Instead of reading large tables of data, users can analyze expenses visually through graphs generated using Python libraries.

In this project, the matplotlib library is used to create charts for expense analysis. The system collects expense data from the MySQL database and displays category-wise expense reports using bar graphs. This makes it easier to compare spending across different categories such as Food, Transport, Shopping, Education, and Entertainment.

Graphs help users identify which category has the highest expenses and where money is being spent unnecessarily. Visual representation also improves the overall presentation and practicality of the project.

Types of Charts Used

1. Bar Graph

The bar graph displays category-wise expenses using vertical bars. Each bar represents a category, while the height of the bar shows the total amount spent.

2. Pie Chart (Optional)

A pie chart can also be used to show the percentage distribution of expenses among different categories.

Python Code for Bar Graph:

```
import matplotlib.pyplot as plt


categories = ['Food', 'Transport', 'Shopping', 'Education', 'Entertainment']

amount = [250, 120, 1500, 800, 300]


plt.bar(categories, amount)
```

```
plt.xlabel("Category")

plt.ylabel("Amount")

plt.title("Category Wise Expenses")


plt.show()
```

Advantages of Data Visualization

- Makes data easy to understand
- Helps in expense analysis
- Improves financial planning
- Provides graphical representation of records
- Enhances project presentation quality

Data visualization is an important part of this project because it converts raw expense data into meaningful graphical insights for better decision-making.

CONCLUSION

The Expense Tracker System is a useful and efficient application developed using Python and MySQL for managing daily expenses digitally. The project helps users store, organize, search, and analyze expense records in a systematic manner. It reduces the difficulties of maintaining manual financial records and provides a faster and more accurate solution.

The system successfully demonstrates the practical implementation of programming concepts such as loops, conditional statements, functions, database connectivity, and data visualization. By using MySQL as the backend database, the project ensures secure and organized storage of expense data. The use of Python libraries like pandas and matplotlib enhances the functionality of the system by providing data analysis and graphical representation features.

Through this project, users can monitor their spending habits, identify unnecessary expenses, and manage their budgets effectively. The graphical analysis through charts and graphs makes expense tracking easier to understand and more user-friendly.

This project also helped in improving programming knowledge, logical thinking, problem-solving ability, and database management skills. Overall, the Expense Tracker System is a practical real-life application that demonstrates how technology can simplify financial management and improve efficiency in everyday life.

BIBLIOGRAPHY

The following books, websites, and online resources were referred to during the development of the Expense Tracker System project. These resources helped in understanding Python programming, MySQL database connectivity, and data visualization techniques.

Books Referred

1. Informatics Practices Class XII Textbook – CBSE
2. Computer Science with Python by Preeti Arora
3. Python Programming for Beginners
4. Database Management System Concepts s

Websites Referred

- Python Official Website
- MySQL Official Website
- Matplotlib Documentation
- Pandas Documentation
- W3Schools Python Tutorial

Purpose of References

These resources were used for:

- Learning Python syntax and programming concepts
- Understanding MySQL database operations
- Connecting Python with MySQL
- Creating graphs and charts using Matplotlib
- Improving coding and project documentation skills

The bibliography section acknowledges all the sources and materials that contributed to the successful completion of the project.